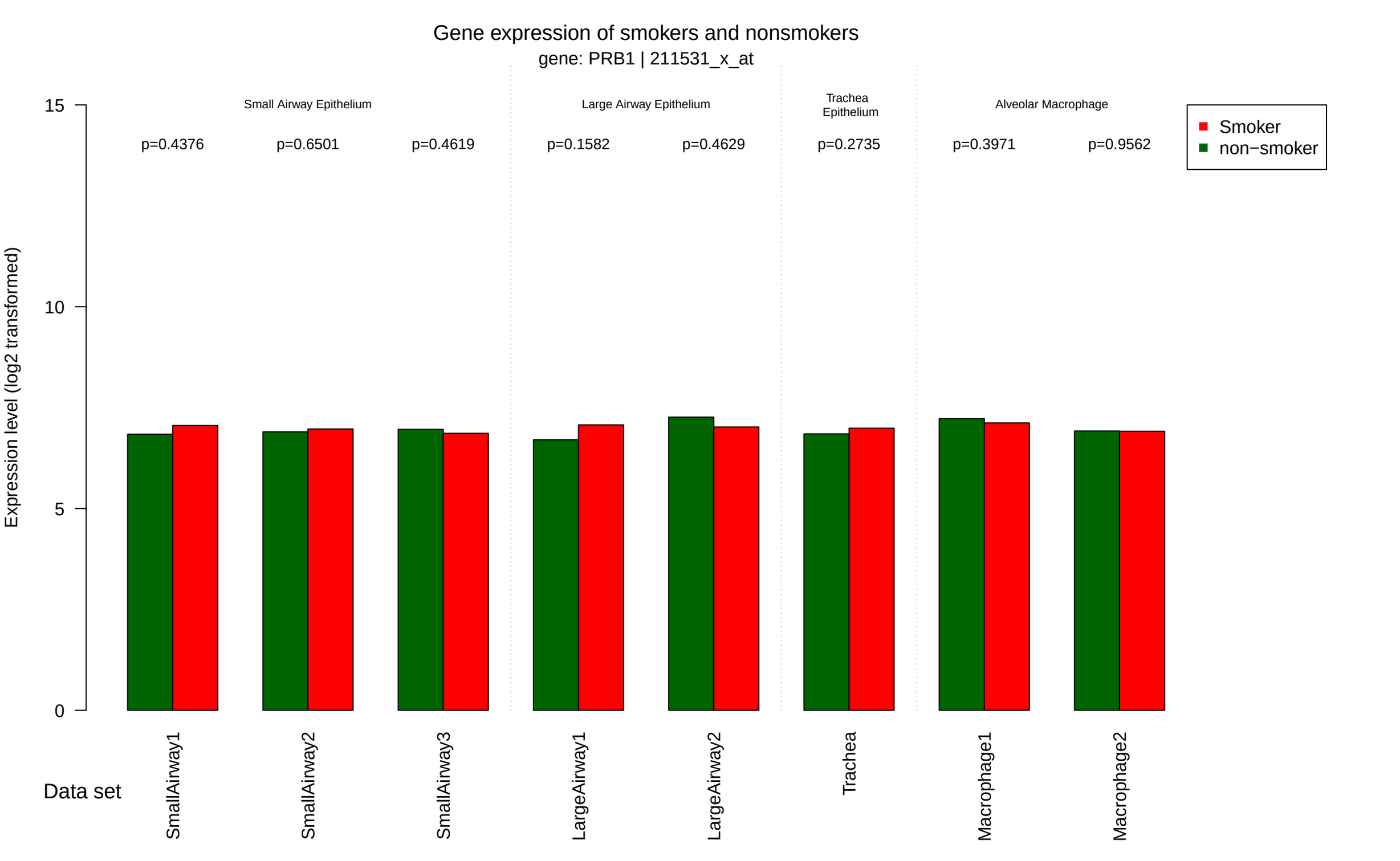
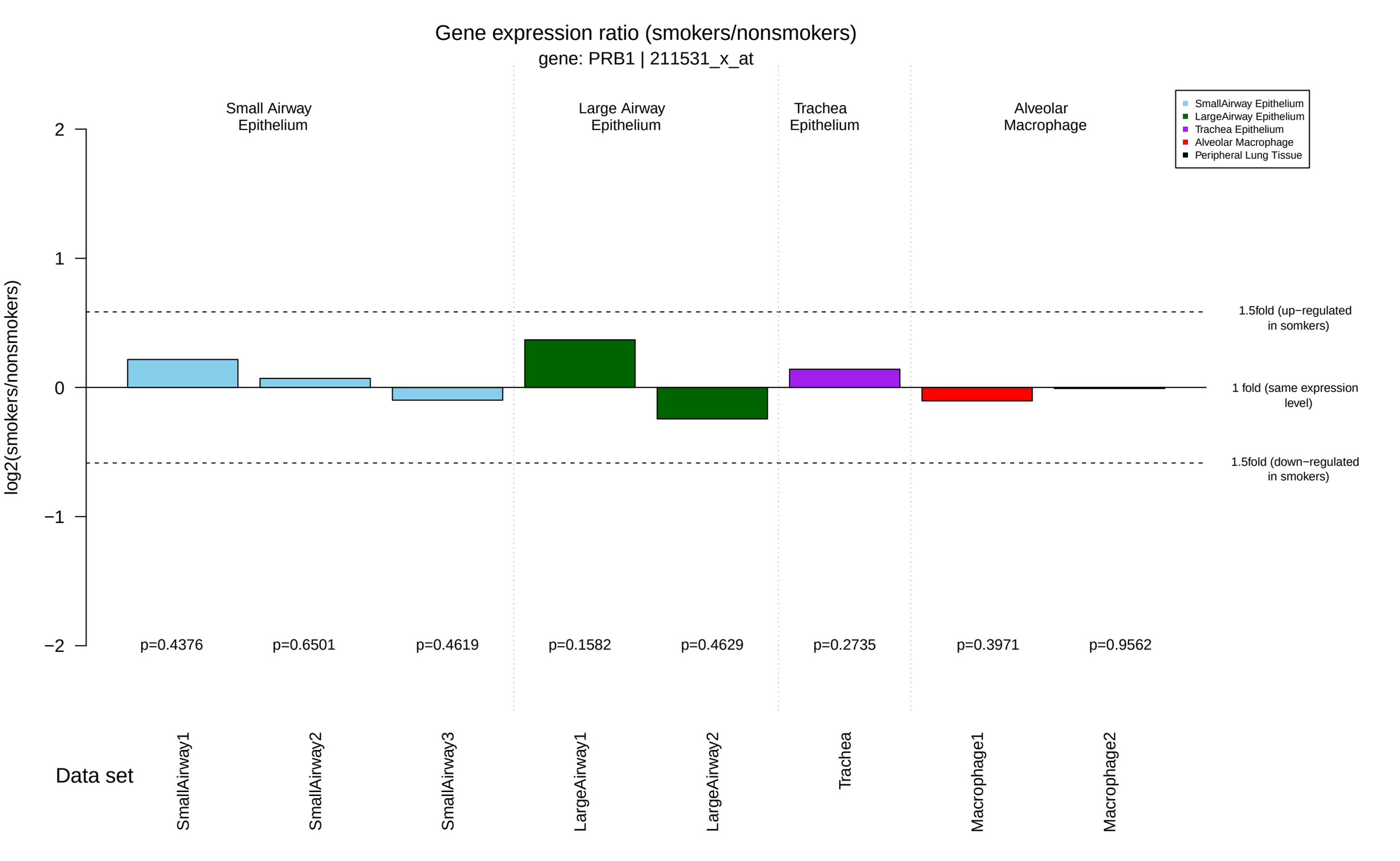
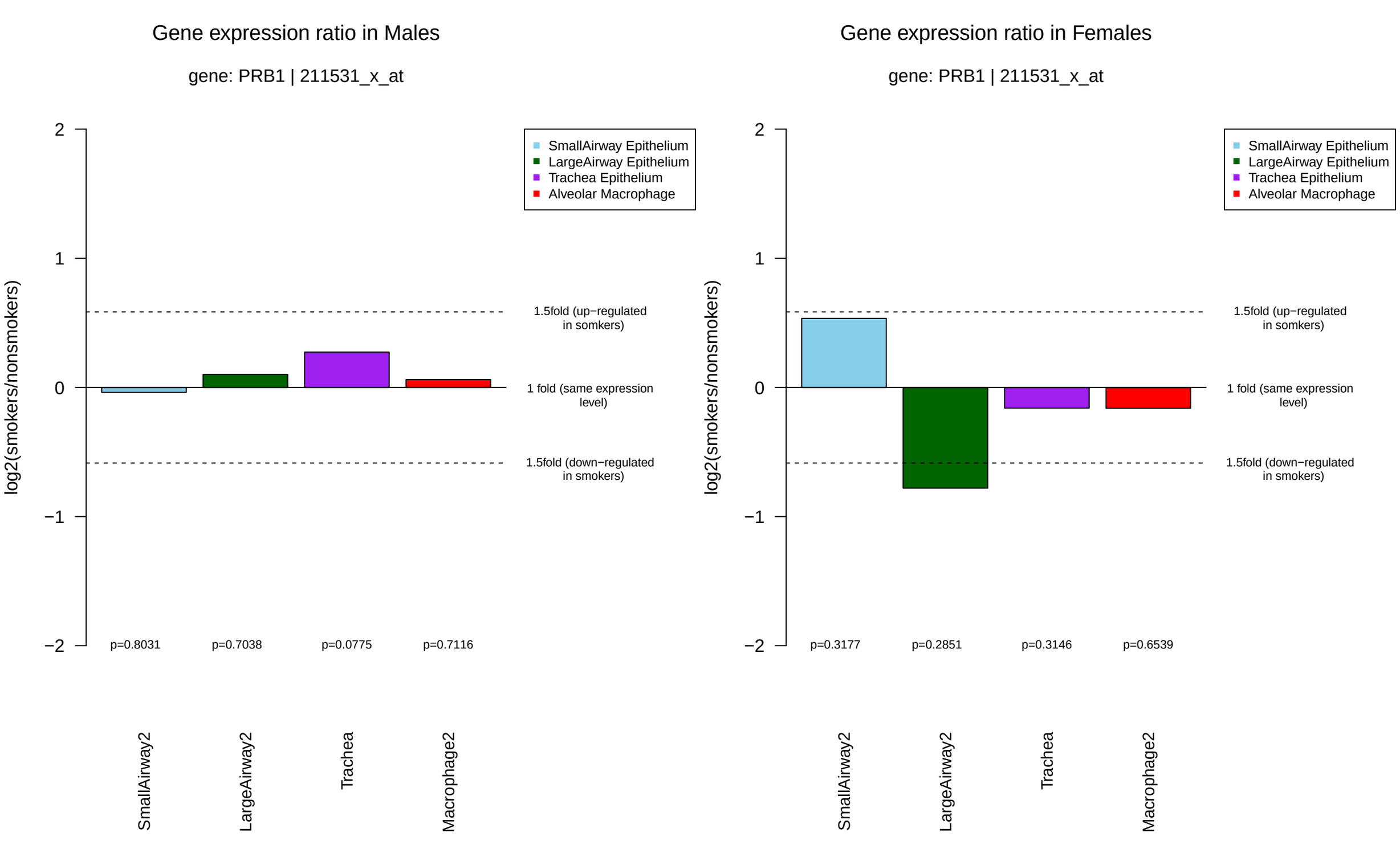
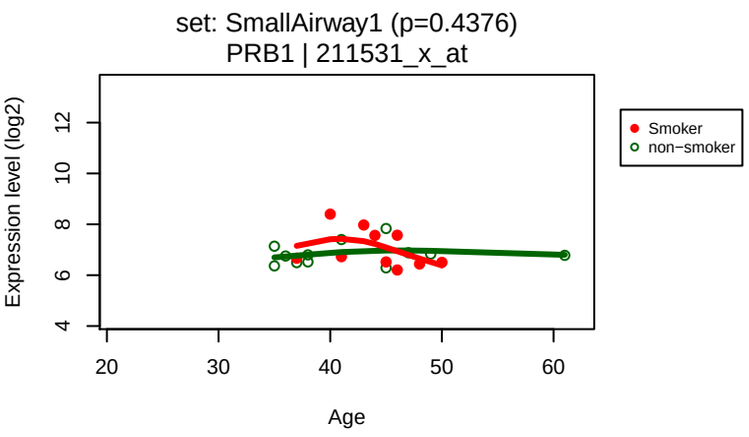
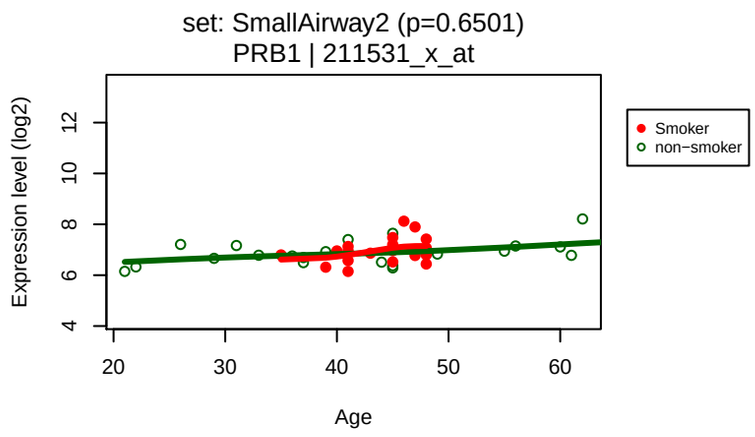
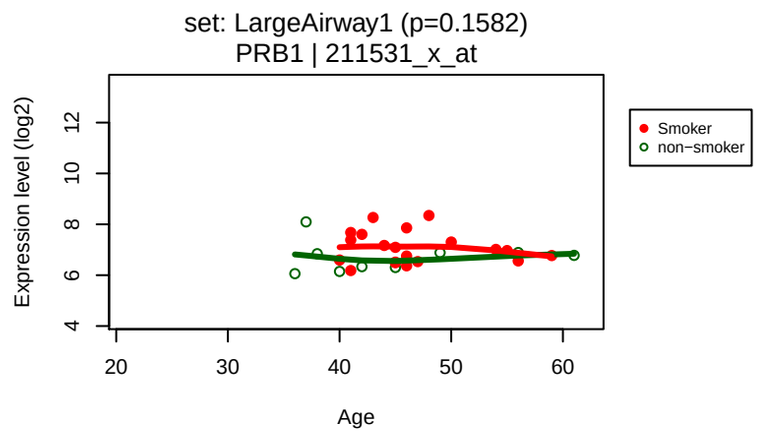

Gene expression of smokers and nonsmokers
gene: PRB1 | 211531_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4376
p=0.6501
p=0.4619
p=0.1582
p=0.4629
p=0.2735
p=0.3971
p=0.9562
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

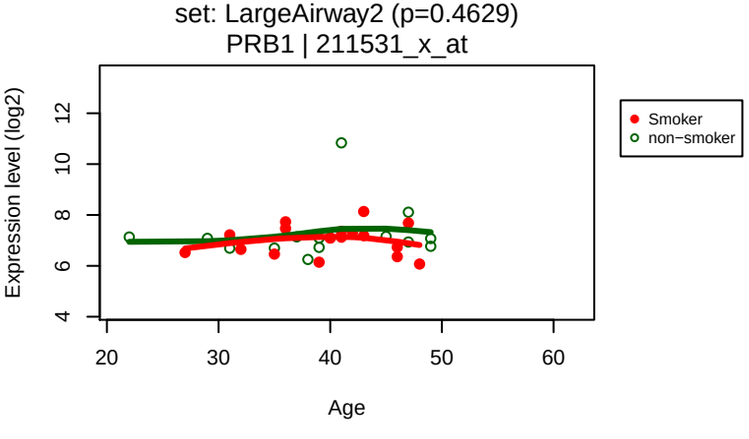

Gene expression ratio (smokers/nonsmokers)
gene: PRB1 | 211531_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4376
p=0.6501
p=0.4619
p=0.1582
p=0.4629
p=0.2735
p=0.3971
p=0.9562
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

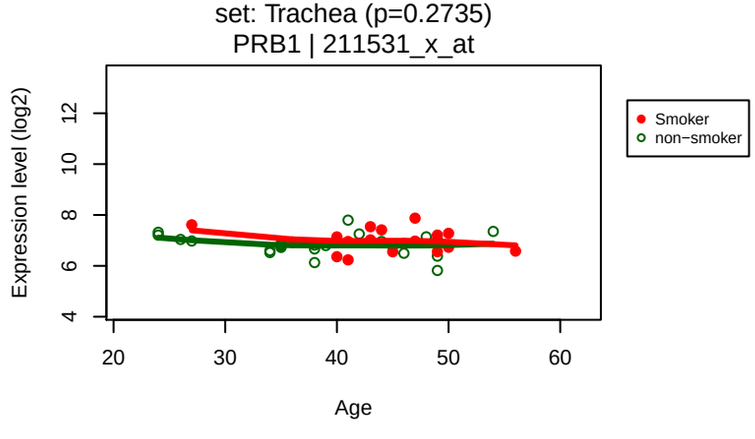

Gene expression ratio in Males
gene: PRB1 | 211531_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8031
p=0.7038
p=0.0775
p=0.7116
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRB1 | 211531_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3177
p=0.2851
p=0.3146
p=0.6539
SmallAirway2
LargeAirway2
Trachea
Macrophage2

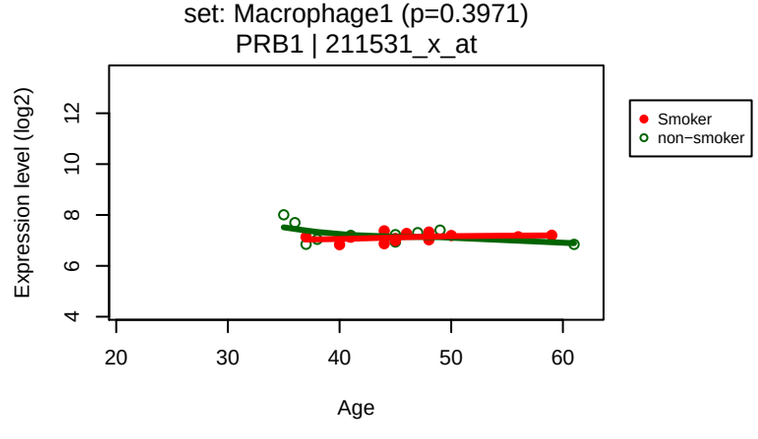

set: SmallAirway1 (p=0.4376)
PRB1 | 211531_x_at
Expression level (log2)
Age
Smoker
non−smoker

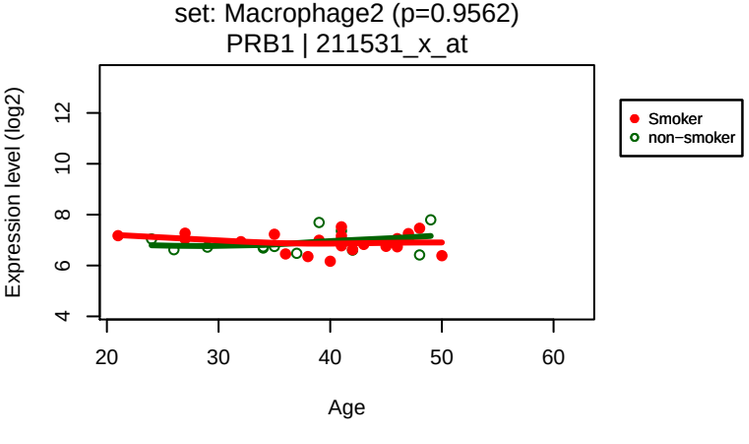

set: SmallAirway2 (p=0.6501)
PRB1 | 211531_x_at
Expression level (log2)
Age
Smoker
non−smoker

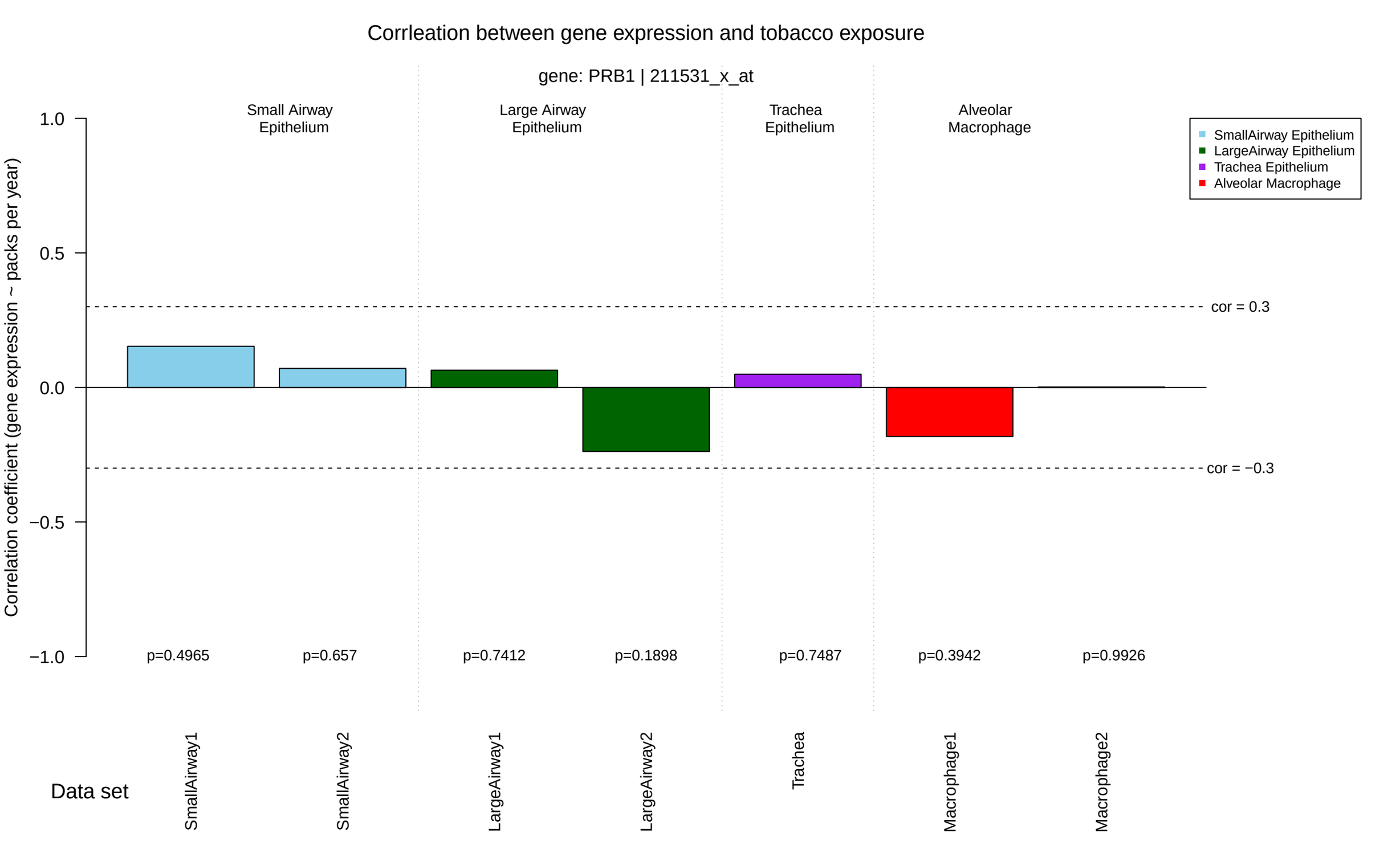

set: LargeAirway1 (p=0.1582)
PRB1 | 211531_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4629)
PRB1 | 211531_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2735)
PRB1 | 211531_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3971)
PRB1 | 211531_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9562)
PRB1 | 211531_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRB1 | 211531_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4965
p=0.657
p=0.7412
p=0.1898
p=0.7487
p=0.3942
p=0.9926
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2